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ML
Project 1 Report on epoch and alpha

I first tested a low epoch and a low alpha value. I found that the hyperparameters were very low (between 1 and 8), and that my training and test losses were both around 5000. The example I provided below shows the inputs for both, as well as the outputs I received.

```
w = train(Xtrain, Ytrain, alpha = 0.001, n_epoch = 100)
```

```
Parameter: [1.80848521 4.47556644 4.603706 7.04561834]  
Train Loss: 4963.9620199457195  
Test Loss: 5232.868297430126
```

Next, I attempted to use a high number of epochs and a low alpha value. I found that the hyperparameters were much higher than in the first test, with a low number of epochs and a low alpha value, ranging from 12 to 46. My training and test losses were also significantly lower than before, both being within the one-thousand range.

```
w = train(Xtrain, Ytrain, alpha = 0.001, n_epoch = 1000)
```

```
Parameter: [12.51533654 30.05272454 36.51306709 45.94906067]  
Train Loss: 1102.0213446265777  
Test Loss: 1223.4043371666112
```

After that, I tested a low number of epochs but a high alpha value. I found that the hyperparameters were almost identical to the low alpha value and high number of epochs, ranging between 12 and 46. As for the training and test losses, they once again sat within the 1000s range. While better than low-epoch and low-alpha values, this was still not the perfect solution.

```
w = train(Xtrain, Ytrain, alpha = 0.01, n_epoch = 100)
```

```
Parameter: [12.54661491 30.1316429 36.59002301 46.0655157 ]  
Train Loss: 1095.2454196436697  
Test Loss: 1216.4037284936414
```

Finally, I tested a high number of epochs with a high alpha value. I found that this was the best solution, with my hyperparameters ranging from 21 to 76 and my training and testing losses below 0.01.

```
w = train(Xtrain, Ytrain, alpha = 0.01, n_epoch = 1000)
Parameter: [21.078077 46.97589201 75.10722798 76.22214929]
Train Loss: 0.0021839380086190346
Test Loss: 0.0028721752802048743
```

In conclusion, I found that when either the number of epochs or the alpha value is higher, the training and test data become more accurate with less loss. However, when testing with an alpha value higher than 0.01 (for example, 0.1 or 1), there seemed to be progressively lower amounts of loss in the training and testing data. When increasing the number of epochs, there was a point at which the loss stopped changing, and only the alpha value would have any effect. So, in my testing, there seemed to be a point where increasing the number of epochs improved loss efficiency, while changing the alpha value had a much greater effect.

```
w = train(Xtrain, Ytrain, alpha = 0.01, n_epoch = 10000)
Parameter: [21.07645025 46.94790286 75.17077209 76.26320954]
Train Loss: 8.371190135585821e-25
Test Loss: 1.1113747745209538e-24
```

```
w = train(Xtrain, Ytrain, alpha = 0.1, n_epoch = 10000)
Parameter: [21.07645025 46.94790286 75.17077209 76.26320954]
Train Loss: 7.103692451515211e-27
Test Loss: 9.525416544453195e-27
```

```
w = train(Xtrain, Ytrain, alpha = 1, n_epoch = 10000)
Parameter: [21.07645025 46.94790286 75.17077209 76.26320954]
Train Loss: 7.85863233781172e-29
Test Loss: 6.446571317466108e-29
```

```
w = train(Xtrain, Ytrain, alpha = 1, n_epoch = 100000)
Parameter: [21.07645025 46.94790286 75.17077209 76.26320954]
Train Loss: 7.85863233781172e-29
Test Loss: 6.446571317466108e-29
```